

Volatility on Stock Exchanges and Exchange Rates Based on Pairwised Alignment

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Abstract

In finance, volatility refers the degree of financial prices fluctuating over time on stock market. It is utilized to quantify a financial instrument in a given particular time period. In decades, empirical researches have various opinions on the impact of exchange rate for security's price. Some scholars are united in their belief that exchange rate movements provide little or no explanation for stock returns; relatively, others argue that security's price are significantly correlated with inflation and money growth. Viewpoints are divergent, it has provoked a great deal of controversy. In this study, we present VOSSA (Volatility On Stock Symbol Alignment), a scaling approach for stock volatility evaluation. Our experiment shows that international exchange rate fluctuations held a relatively large fraction of variation in Taiwan stock market. Comparing with statistical inference approach, our method provides an alternative conception on its efficiency. This numerical approach offers a practical direction on the impact of exchange rate for security's price comparison.

1 Introduction

For the past few decades, financial system has a close relationship with the mathematics, which is concerned with much of the underlying theory. Financial engineering, also called computational finance, relies heavily on mathematically calculating to estimate various financial subjects. It involving market investment, trading economics, risk management, banking, and forecasting [1], the field applies engineering methodologies to problems in finance and employs financial theory and applied mathematics.

Estimating the relationship between security's price and exchange rates is essential for the following reasons: First, it may affect decisions about monetary and fiscal policy. One research indicates that a booming stock market has a positive effect on aggregate demand [2]. Second, the link between the two markets may be used to predict the path of the exchange rate. This will benefit multinational corporations in hedging transaction exposure by currency forward contracts, eliminating its exchange risk exposure [3]. Third, a currency that increases relative to its portfolio can outperform a large gain in a retreating currency. For evaluative purposes, the investor should analyze the various asset classes in relation currency [4].

Determining the effects of these model variables on stock prices and investment decisions is always challenging since interest rate, inflation, exchange rates, money supply, and stock prices are involved affectingly [5]. However, under certain circumstances, these models based on current financial theories cannot explain long-observed features of the implied volatility surface [6]. The direction of causality still remains unresolved in both theory and empirics. Therefore, considering this subject from an alternative point of view, our motivation is to develop a fast and facile method to estimate the impact of exchange rate for security returns.

2 Methods

In this section, to explain our proposed algorithm, we start with a broad overview of the problem by VOSSA procedure. This method can divide into three phases: First, financial raw data set are normalized using Z-score process; second, volatilities are categorized into discrete state by specific indexing; third, sequences of price movement are aligned to find the best-matching piece-

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wise. Since a time series volatility model has transferred into a sequence data set, it becomes the problem of multiple sequence alignment which lead to NP-complete combinatorial optimization problems. VOSSA applies daily data of Taiwan security's price and exchange rate of Taiwan, Japan, Korea, and U.S., investigating the co-movement among stock market from 2007 to 2011.

2.1 Problem Formations

Since variables have an great impact on daily stock prices, especially the continuing world trade and capital movements, exchange rates become one of the main determinants of business profitability and equity prices [7]. Thus, understanding this relationship helps investors for hedging and diversifying their financial portfolio. In this study, a method is proposed to detect volatilities between stock returns and exchange rates, aiming on a symbolic analysis of time series economic model.

Suppose $X = \{x_1, \dots, x_t\}$ is a collection of stock return patterns within a time period t and $Y = \{y_1, \dots, y_t\}$ is a collection of exchange rate patterns with the same time period. Let $L = \{l_1, \dots, l_k\}$ be a scale or non-scale intervals, where, k is the number of symbols. In addition, the calculation of volatility V can be formulated as follows, where r is raw data of exchange rate or security's price.

$$v_i = \frac{r_i - r_{i-1}}{r_{i-1}} \quad (1)$$

After the normalization and symbolization, vector of X has transferred to X_V then X_S , as similar as Y_V then Y_S . Since the problem has become a sequence alignment problem, the goal is to find a maximal regions of similarity among given patterns. Hence, in the following section, denoting more pages to discuss the procedure of VOSSA is essential.

2.2 Symbolization

It is desirable to describe normalization before moving on to the main task. An exchange rate is regarded as the value of one countrys currency in terms of another currency [8]; therefore, instead of converting the currency, we simply bringing them to a common scale, allowing unlike distributions to be compared in a standard manner. Z-score—a common statistical way of standardizing data

on one scale so a comparison can be done. By definition, the formula of Z-score can be written as follows:

$$\mathbf{Z} = \{z_1, z_2, \dots, z_k\}, \quad z_i = \frac{x_i - \mu}{\sigma} \quad (2)$$

where k is the length of time series, x is stock price or exchange rate which needs to be adjust, μ is the mean of X , and σ is the standard deviation of X .

We now consider the symbolization procedure. Due to the limitation of Z-score, the scale range would be ± 4 . Suppose t is the time series of stock returns with the range scale k , a countable set of delimiter values are:

$$-4 \leq l_1 \leq l_2 \leq \dots \leq l_{k-1} \leq l_k \leq 4 \quad (3)$$

which represent the intervals or the number of symbols k and k needs to be an even number. For example, if $k = 2$, which means $|L| = k + 1 = 3$; therefore, two binary symbols are required when $z_i > 0$ or $z_i < 0$. It should be noticed that the process of normalization is after volatility measuring. This helps the simplicity on delimiter values determination. It seems reasonable to suppose that intervals of delimiters are equal, in Formula 3, that is $|l_1 - l_2| = |l_{k-2} - l_{k-1}| = \dots = |l_{k-1} - l_k|$. In view of the property of Z-score, let us consider an inconstant interval on symbolization.

Obviously, in Z-score, 68% of data are concentrated between ± 1 ; therefore, intervals of delimiter values should be inconstant and advancing. We found that the rule of **Fibonacci Sequence** holds in this case. By definition, each subsequent Fibonacci number is the sum of the previous two. In short, there are $k/2$ intervals and $(k/2) - 1$ parameters between 0 to 4. Besides, in order to satisfy the growing ratios, we ignore the first number in Fibonacci sequence.

2.3 Sequence Alignment

Once the symbolic representation of financial data set is decided, strings of symbols can be matched by sequence comparison. There exist many extensive research on sequence alignment topic, one of the most notable method is BLAST [9], calculating the statistical significance for each sequence alignment result. Hence, BLAST is adapted as a tool in VOSSA. The only issue needs to be mentioned here is scoring function. Scoring functions, defined by mathematic formulas, provide a score to each matching document for

Table 1: The scoring matrix used in this study.

-	A	R	N	D	C	Q	E	G	H	I	L	K
A	21	13	8	5	3	1	-1	-3	-5	-8	-13	-21
R	-	21	13	8	5	3	-3	-5	-8	-13	-21	-34
N	-	-	21	13	8	5	-5	-8	-13	-21	-34	-55
D	-	-	-	21	13	8	-8	-13	-21	-34	-55	-89
C	-	-	-	-	21	13	-13	-21	-34	-55	-89	-89
Q	-	-	-	-	-	21	-21	-34	-55	-89	-89	-89
E	-	-	-	-	-	-	21	13	8	5	3	1
G	-	-	-	-	-	-	-	21	13	8	5	3
H	-	-	-	-	-	-	-	-	21	13	8	5
I	-	-	-	-	-	-	-	-	-	21	13	8
L	-	-	-	-	-	-	-	-	-	-	21	13
K	-	-	-	-	-	-	-	-	-	-	-	21

a query. The choice of a scoring function that reflects biological or statistical observations about known sequences is important for producing good alignments [10].

3 Results

In order to demonstrate our method, we examine the volatility data set of exchange rate and security's price in Taiwan from 2007 to 2011. In this study, we relax some conditions to observe its validity on financial problems. Let k be senary and using specific scoring function in Table 1, $L = [-4, -2.5, -1.5, 0, 1.5, 2.5, 4]$, $|k| = 10$. In addition, the value in scoring matrix is comprehended by Fibonacci sequence.

Table 2 gives comparative values obtained by VOSSA procedure. In this table, five exchange rates are displayed their relationship with stock returns. *Commu* for communication sector, *Compu* for computer sector, *Dist* for distribution of 3Cs products industry, *Elec* for electronics sector, *Info* for information service sector, *Opti* for optical sector, *Other* for other electronics, *Peri* for PC and peripherals, and *Semi* for semiconductors. B Bit score in each column indicates the highest alignment results. Q and S are "Query position" and "Subject position", respectively, indicating the date number from 2007 to 2011 of stock market.

It is easy to observe that the communication sector is likely related with most exchange currency. This may explain the growth of network and communication appliances, built on WiMAX, 3G network, embedded system, Android operating system, broadband, and wireless industries in Taiwan from 2007 to 2011. Besides, semiconductor sector is also related from 2007 to the first half of 2009. From the point of foreign exchange currency, U.S. Dollar to Taiwan Dollar is significant associated with most stock sectors. Results suggest that correlation between foreign market varies significant extent with time; in addition, exchange rate movements provide a large explanation in investment decisions.

4 Conclusion

In this study, we have developed a symbolic analysis approach—VOSSA, investing the relationship between exchange rate for security's price. Unlike traditional financial econometric models and applications, there have been many empirical and theoretical studies to define the direction of causality between these two financial variables. Instead, VOSSA allows financial data aligning its subsets in time series to measure their degree of similarity, capable to locate interesting ranged features on stock returns. This symbolic analysis approach offers a practical direction for detecting relationship between exchange rate and security's price.

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Table 2: The similarity of exchange rate and security's price in Taiwan from 2007 to 2011.

Sector		YEN USD	WON USD	TWD USD	TWD YEN	TWD WON
Commu	B	355	247	277	368	-
	Q	10 - 978	13 - 974	6 - 923	6 - 977	-
	S	6 - 961	13 - 959	2 - 961	8 - 970	-
Compu	B	313	-	265	-	-
	Q	18 - 714	-	21-905	-	-
	S	26 - 816	-	21-928	-	-
Dist	B	221	-	216	235	-
	Q	22 - 906	-	32 - 905	22 - 906	-
	S	27 - 919	-	34 - 977	21 - 970	-
Elec	B	316	-	249	316	-
	Q	7 - 714	-	4 - 714	8 - 714	-
	S	27 - 824	-	2 - 790	21 - 861	-
Info	B	-	209	258	-	244
	Q	-	107 - 706	34 -973	-	28 - 969
	S	-	78 - 811	6 - 974	-	27 - 966
Opti	B	392	-	329	-	-
	Q	22 - 705	-	23 - 705	-	-
	S	27 - 761	-	24 - 906	-	-
Other	B	-	-	-	365	-
	Q	-	-	-	19 - 715	-
	S	-	-	-	19 - 841	-
Peri	B	218	-	-	235	-
	Q	25 - 906	-	-	25 - 637	-
	S	47 - 924	-	-	23 - 765	-
Semi	B	252	-	211	225	225
	Q	19 - 474	-	20 - 474	19 - 474	19 - 474
	S	27 - 555	-	6 - 574	22 - 521	33 - 557

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